

Misinformation Detection Using Machine Learning Models

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1. Introduction

This project aims to explore and model misinformation patterns using machine learning techniques.

The main objectives are:

1. To understand the data distribution of misinformation.
2. To preprocess and standardize the dataset for model training.
3. To build and evaluate predictive models (Logistic Regression and CART).
4. To identify the most influential indicators of misinformation.

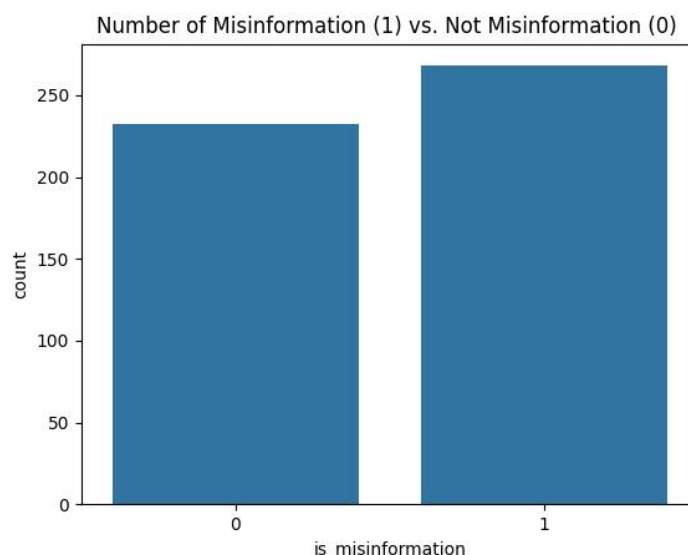
By combining data exploration and supervised learning, this project demonstrates how machine learning can be applied to detect misinformation efficiently.

2. Data Preprocessing and Exploration

2.1 Exploratory Data Analysis (EDA)

The first step involves understanding the data composition and identifying key trends.

- **Proportion of Misinformation:** The dataset shows a higher number of misinformation records compared to non-misinformation ones, indicating that misinformation occurs more frequently.



Graph 1 Proportion of *is_misinformation*

- **Categorical Features:** Twitter contributes the highest number of records (25.8% of total, 29.1% of misinformation). Germany and Los Angeles are the most common country and city respectively.

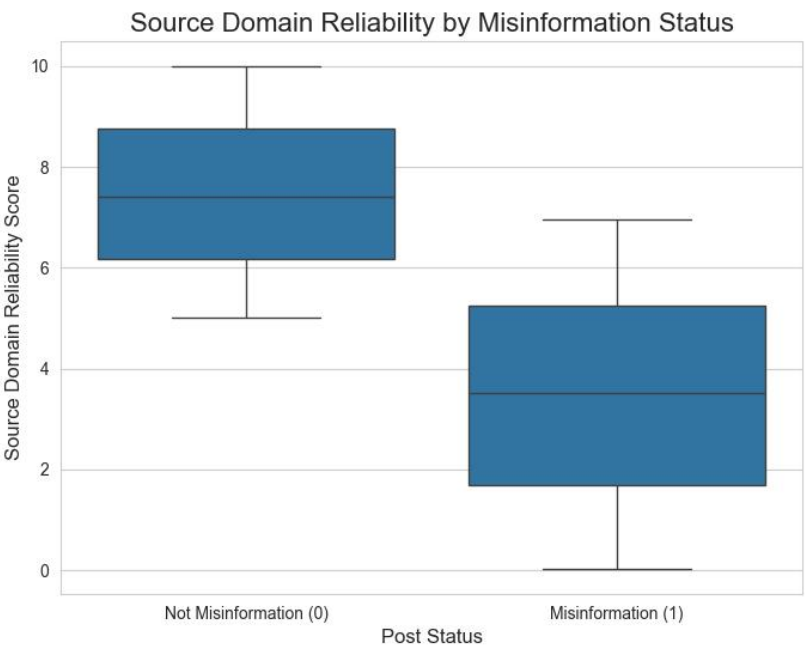
	id	platform	timestamp	date	time	month	weekday	country	city	timezone	author_id
count	500	500	500	500	500	500	500	500	500	500	500
unique		4	500	340	422	12	7	5	15	5	488
top		Twitter	2024/3/6 10:01	28 06 2024	22:49:00	June	Friday	Germany	Los Angeles	CET	A5155
freq		129	1	5	4	65	83	113	51	113	2

Table 1 summary of categorical data in total data set

	id	platform	timestamp	date	time	month	weekday	country	city	timezone	author_id
count	268	268	268	268	268	268	268	268	268	268	268
unique		4	268	216	246	12	7	5	15	5	265
top		Twitter	2024/3/6 10:01	29 08 2024	22:49:00	May	Friday	Germany	Los Angeles	CET	A2600
freq		78	1	3	3	34	48	60	30	60	2

Table 2 summary of categorical data in data set where is_misinformation is true

- **Source Domain Reliability:** Non-misinformation data has higher reliability scores, indicating this feature’s predictive importance.



Graph 2 Source Domain Reliability by Misinformation Status

3. Data Cleaning and Standardization

Before applying machine learning models, several preprocessing steps were conducted:

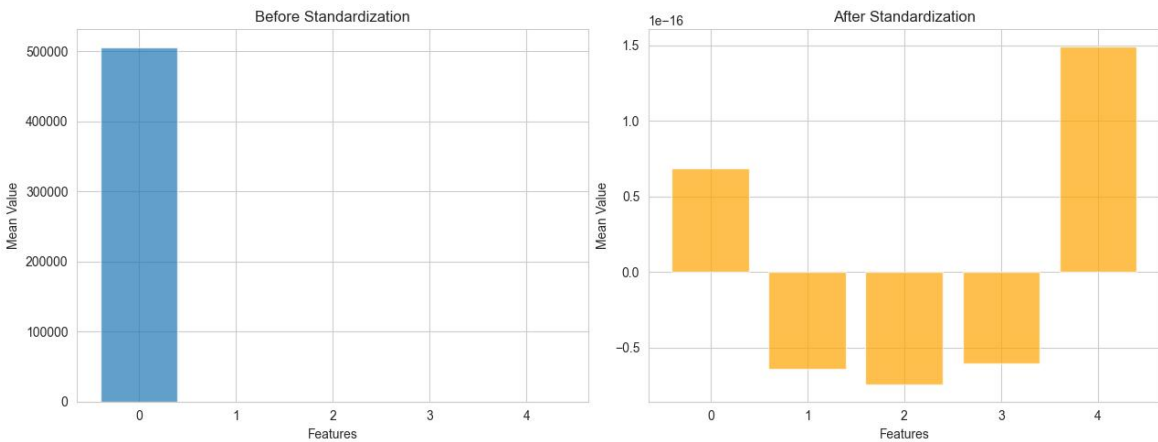
1. **Missing Value and Duplication Check:** No missing or duplicated data was found.

	id	platform	timestamp	date	time	month	weekday	country	city	timezone	author_id	author_followers	author_verified
missing_count	0	0	0	0	0	0	0	0	0	0	0	0	0
missing_percent	0	0	0	0	0	0	0	0	0	0	0	0	0
	text_length	token_count	readability_score	num_urls	num_mentions	num_hashtags	sentiment_score	toxicity_score	detected_synthetic_score	external_factchecks_count	source_domain_reliability	engagement	is_misinformation
missing_count	0	0	0	0	0	0	0	0	0	0	0	0	0
missing_percent	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 3 Missing Value Check

2. **Feature Selection:** Irrelevant columns such as id, date, time, and timestamp were removed.

3. **Data Standardization:** Standardization was performed to align variable scales and improve model efficiency.



Graph 3 Data Standardization

These steps ensured that the dataset was clean, balanced, and ready for model training.

4. Machine Learning Model Development

Two supervised learning models were trained and tested to predict misinformation.

4.1 Logistic Regression:

1. A statistical model estimating the probability that a given input belongs to the misinformation class.

2. Categorical variables were one-hot encoded to enhance interpretability.

3.The categorical variables were converted into dummy variables (one-hot encoding) to enhance interpretability.

4.2 CART (Classification and Regression Tree):

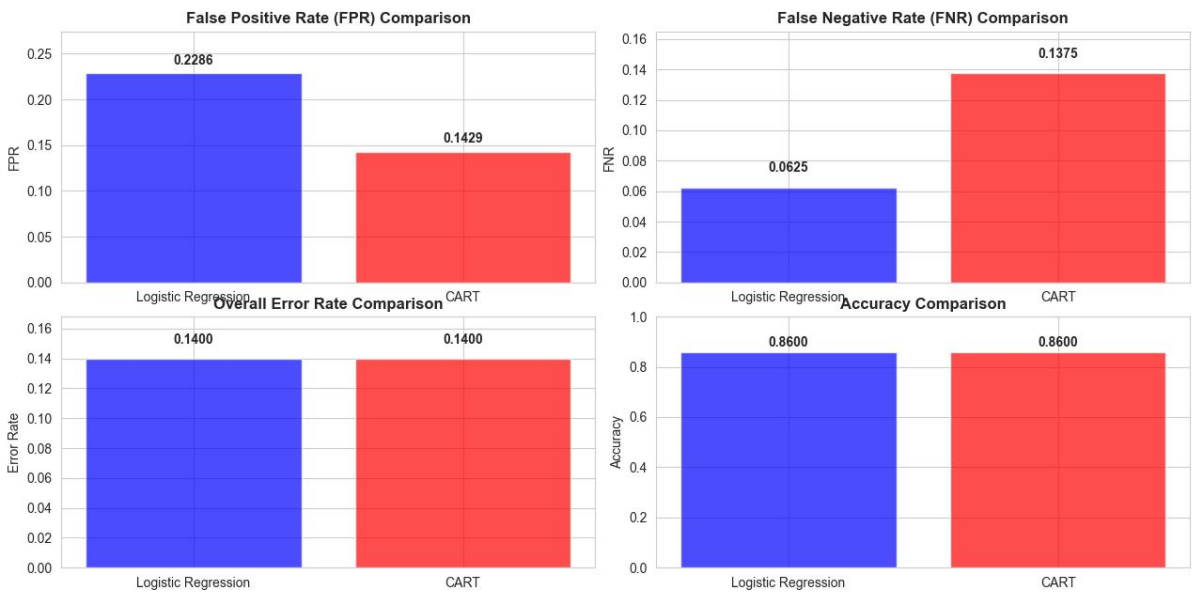
- 1.A decision tree-based algorithm that classifies data by learning simple if-else rules.
- 2.It captures non-linear feature interactions.

5. Model Evaluation and Comparison

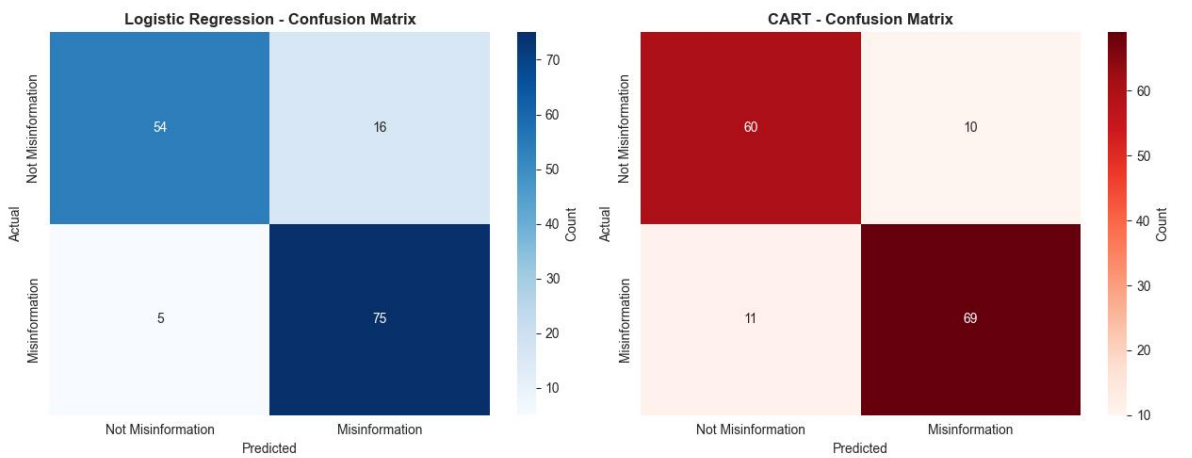
Model performance was evaluated using accuracy, error rate, and confusion matrix.

Model	Model Complexity	False Positive Rate	False Negative Rate	Overall Error	Accuracy
Logistic Regression	543 X variables	0.2286	0.0625	0.14	0.86
CART	16 terminal nodes	0.1429	0.1375	0.14	0.86

Table 4 Model comparison between Logistics Regression and CART



Graph 4 Model comparison between Logistics Regression and CART



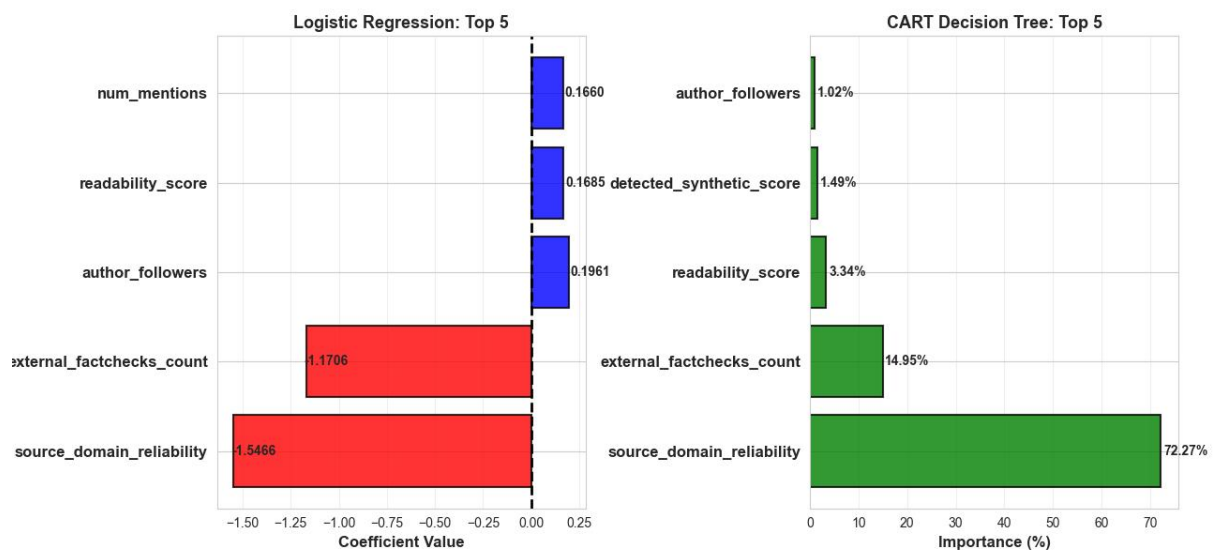
Graph 5 Confusion Matrix of Logistics Regression and CART

Both models achieved similar accuracy (86%), but Logistic Regression performed better in minimizing false negatives—important for misinformation detection, as missing a misinformation case is more costly than a false alarm.

6. Feature Importance Analysis

Feature importance analysis helps to identify which variables have the strongest predictive power.

- **Logistic Regression:** Source Domain Reliability and External Fact Checks Count are the top predictors.
- **CART:** Feature importance ranking confirms the same two variables as dominant.



Graph 6 Most significant Indicator in Logistic Regression and CART

These results highlight the critical role of source credibility and external validation in misinformation detection.

7. Discussion

The findings show that:

- **Source Domain Reliability** strongly correlates with misinformation likelihood.
- **External Fact Checks Count** supports this pattern.
- **Logistic Regression** is more robust for this dataset due to its interpretability and lower false negative rate.
- Identical 86% accuracy across both models may be due to small dataset size or lack of data diversity.

8. Technical Summary

Programming Language: Python 3.11

Key Libraries: pandas, scikit-learn, seaborn, matplotlib, numpy

Algorithms Used: Logistic Regression, DecisionTreeClassifier (CART)

Data Split: 70% training / 30% testing, stratified sampling

Model Metrics: Accuracy, False Positive Rate, False Negative Rate, Overall Error

Best Model: Logistic Regression (Accuracy = 0.86, Low False Negative Rate)

Feature Engineering: One-Hot Encoding for categorical variables, StandardScaler for normalization

Visualization Tools: Seaborn and Matplotlib for feature distribution, model comparison, and importance plots

9. Conclusion

This project demonstrates the application of machine learning in misinformation detection.

By preprocessing the dataset, applying Logistic Regression and CART models, and analyzing feature importance,

we identified that Source Domain Reliability and External Fact Checks Count are the strongest indicators of misinformation.

Future work can include expanding the dataset and incorporating ensemble methods (e.g., Random Forest, XGBoost) to enhance prediction accuracy and generalization.

Answer to Q5:

Through the general data exploration, it is notable that in the data set, the misinformation have higher proportion than not misinformation, this illustrates that if the data set is unbiased, the misinformation happens more often than not misinformation. Also, Twitter is the platform that appears most in the whole data set and misinformation. Germany and Los Angeles are the Country and the city have most data in the data set and misinformation respectively. Moreover, source domain reliability is a strong indicator of misinformation.

By doing CART and Logistic Regression to the original data set, it shows that two models have similar accuracy, both 86%. However, since Logistic Regression have lower False Negative rate. The fact of 86% accuracy for both model is not normal, this could potentially caused by the size of data set is too small.

Additionally, significant indicators analysis has been conducted, it has been found that both source domain reliability and external fact checks count are strong indicators for both CART and Linear Regression Model. This demonstrates the high correlation between source domain reliability and external fact checks count with misinformation.

Appendix: List of GenAI Prompts and Outputs